

Leetcode 981

set(1, 10, 1) ←

set(5, 15, 6) ←

set(1, 20, 10) ←

get(1, 1) → 10 ✓

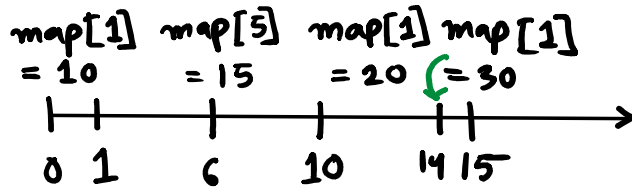
get(1, 5) → 10

get(1, 10) → 20

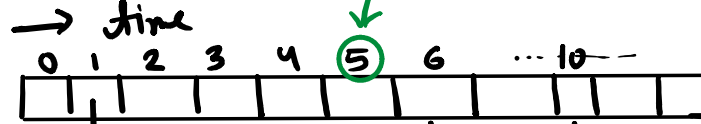
get(1, 15) → 20

set(1, 30, 14) ←

get(1, 15) → 30



1. Set will happen in increasing order



Key-Value ↓ [1, 10]

[5, 15] [1, 20]

Key
①
5

(timestamp, value)
→ [1, 10], [10, 20], [14, 30], ...
[6, 15], ... ↑

Time

1 → [1, 10]

0 → []

2 → [1, 10]

10 → [10, 20]

~~① Binary Search~~

~~② Lower Bound~~

✓ ③ Upper Bound - 1

~~④ Something Else~~

668. Kth Smallest Number in Multiplication Table

FFF ... FTTT
TTT ... TFFF

	1	2	3	4	5	
1	1	2	3	4	5	5
2	2	4	6	8	10	2
3	3	6	9	12	15	1
4	4	8	12	16	20	1

(Binary Search on the answer)

⑤

m = 4

n = 5

5, 5
4

K = 6
4

→ 12

4	4	8	12	16	20	1
---	---	---	----	----	----	---

1 2 2 3 3 4 4 6 6 9

$\boxed{K=6} \rightarrow \boxed{12}$

$$\text{low} = \underline{1}, \text{high} = \underline{20}, \text{mid} = \frac{20+1}{2} = \underline{10}$$

$$\text{low} = 1, \text{high} = 9, \text{mid} = 5$$

15 6

count $\geq K$ $\rightarrow$ Can be a valid answer
 But, also search for the smaller values

count $< K \rightarrow$ Can not be a valid answer
 Search for bigger values

$$T.C: O(\log(m \cdot n) \cdot m)$$

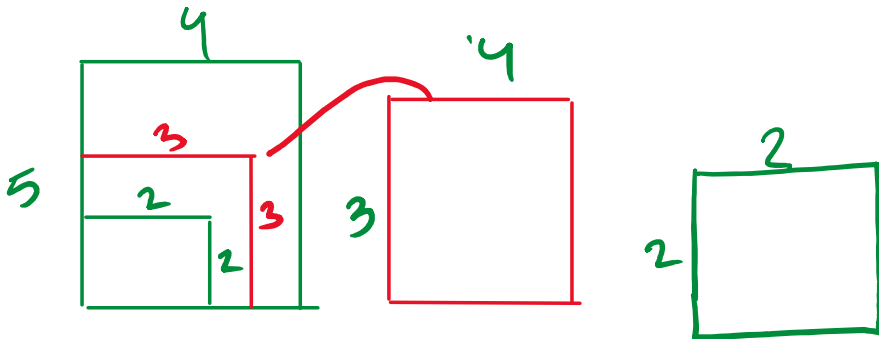
$$[6, \overset{+6}{\curvearrowright} 12, \overset{+6}{\curvearrowright} 18, \overset{+6}{\curvearrowright} 24, \boxed{\overset{+6}{\curvearrowright} 30}, \overset{+6}{\curvearrowright} 36]$$

$$\begin{aligned} 30 &\rightarrow \frac{30}{6} = 5 \\ 24 &\rightarrow \frac{24}{6} = 4 \\ 34 &\rightarrow \frac{34}{6} = 5 \\ 40 &\rightarrow \frac{40}{6} = 6 \\ 22 &\rightarrow \frac{36}{6} = 6 \end{aligned}$$

$$36 \rightarrow \frac{36}{6} = 6$$

$$42 \rightarrow \min\left(6, \frac{42}{\frac{6}{7}}\right) = 6$$

Leetcode 354 - Russian Dolls Envelopes



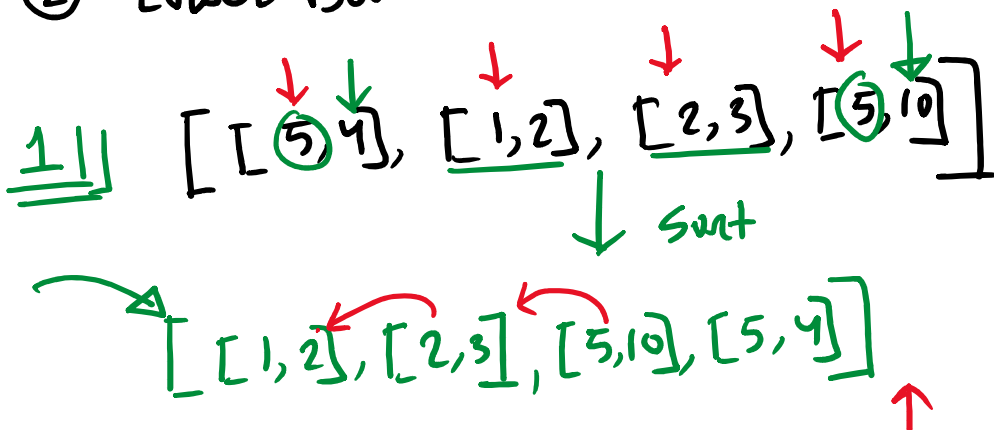
DP $\rightarrow$ LIS (Longest Increasing Sub-seq)

$$O(N^2) \sim O(N \log N)$$

Solution

① Sorting the envelopes

② Lower Bound



2

0	1	2	3	4	5
0	2	3	4	∞	///

↑

← lowerBound

$$\text{answer} = [1, 2, 3, 3] \xrightarrow{\text{max}} = \boxed{3}$$

$$[[2, 3], [5, 2], [5, 7], [5, 5]] \rightarrow [[2, 3], [5, 5], [5, 7], [5, 12]]$$

0	1	2	3	4	5
0	3	5	7	12	///

$$\text{ans} = \min[1, 2, 3, 7] = \boxed{4}$$

$$[[2, 11], [5, 12], [5, 7], [5, 5], [5, 13]]$$

lower bound

0	1	2	3	4	5
0	5	12	∞	∞	///

$$\text{ans} = [1, 2, 1, 1, 3] = \boxed{3}$$